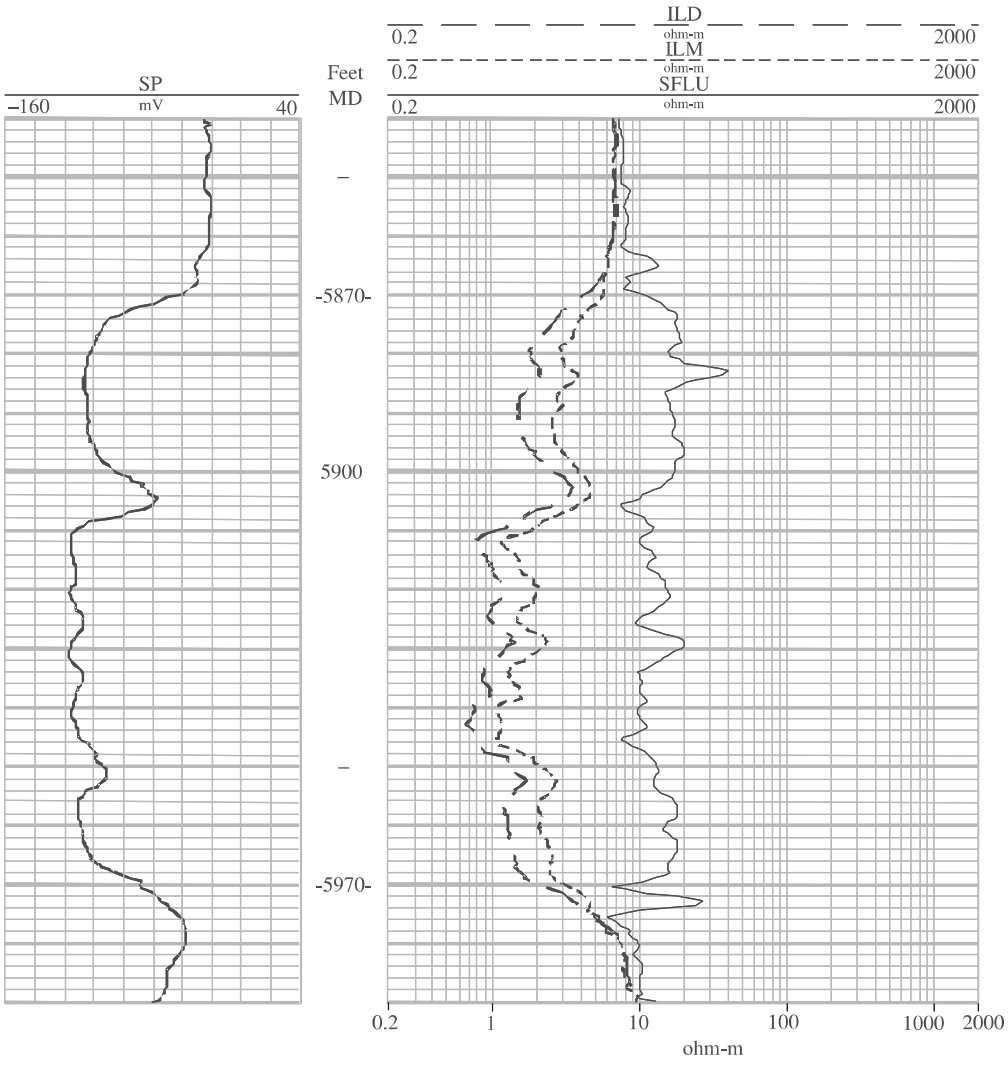


Figure 1.6. Example of dual induction log curves through a water-bearing zone.

Given: the drilling mud is freshwater based ($R_{mf} \geq 3R_w$).

Where freshwater drilling muds invade a water-bearing formation ($S_w > 60\%$), there is high resistivity in the flushed zone (R_{xo}), a lesser resistivity in the invaded zone (R_i), and a low resistivity in the uninvaded zone (R_t).

See Figure 1.4 for review. (Figure 1.8 shows the response of these resistivity curves in a hydrocarbon-bearing zone.)

Compare the three curves on the right side of the log (tracks 2 and 3). Resistivity increases from left to right. A key for reading this logarithmic resistivity scale is shown at the bottom of the log. Depth scale is in feet with each vertical increment equal to 2 ft.

Log curve ILD:

Deep induction log resistivity curves usually measure true formation resistivity (R_t), the resistivity of the formation beyond the outer boundary of the invaded zone. In water-bearing zones (in this case from 5870 to 5970 ft), the curve reads a low resistivity because the pores of the formation are saturated with low resistivity connate water (R_w).

Log curve ILM:

Medium induction log resistivity curves measure the resistivity of the invaded zone (R_i). In a water-bearing formation, the curve reads a resistivity between R_t and R_{xo} because the fluid in the formation is a mixture of formation water (R_w) and mud filtrate (R_{mf}).

Log curve SFLU:

Spherically focused log resistivity curves measure the resistivity of the flushed zone (R_{xo}). In a water-bearing zone, the curve reads a high resistivity because freshwater mud filtrate (R_{mf}) has a high resistivity. The SFL pictured here records a greater resistivity than either the deep (ILD) or medium (ILM) induction curves.

Fluid in the formation is mainly Rmf